

Topics to discuss

- Simpson's $3/8$ Rule
- Numerical Problem
- Homework Problem

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Simpson's $\frac{3}{8}$ Rule :

It works by approximating the function using cubic (3rd degree) curves over three equal sub-intervals at a time (instead of straight line) which usually gives more accurate results than the trapezoidal rule.

Let, $h = \frac{x_n - x_0}{n}$ or $\frac{b-a}{n}$, where n must be multiple of 3.

The formula is :

$$\int_{x_0}^{x_n} y dx = \frac{3h}{8} \left[y_0 + y_n + 3(y_1 + y_2 + y_4 + y_5 + \dots + y_{n-1}) + 2(y_3 + y_6 + y_9 + \dots + y_{n-3}) \right]$$

Numerical Problem :

Using Simpson's 3/8 rule, evaluate approximately

$$\int_0^{1.2} e^{x^2} dx \text{ by taking } n=6.$$

Solution, let, $y = f(x) = e^{x^2}$

$$\therefore \text{Step length, } h = \frac{b-a}{n} = \frac{x_n - x_0}{n} = \frac{1.2 - 0}{6} = 0.2$$

$$\text{So, } x_0 = 0$$

$$x_1 = x_0 + h = 0 + 0.2 = 0.2$$

$$x_2 = x_0 + 2h = 0 + 2 \times 0.2 = 0.4$$

$$\vdots$$

$$x_n = 1.2$$

We have, $f(x) = e^{x^2}$

x	0	0.2	0.4	0.6	0.8	1.0	1.2
x^2	0	0.04	0.16	0.36	0.64	1.00	1.44
e^{x^2}	1.00000	1.040811	1.173511	1.433329	1.896481	2.718282	4.220696

According to Simpson's $3/8$ rule,

$$\begin{aligned}\int_0^{1.2} f(x) dx &= \frac{3h}{8} [y_0 + y_6 + 3(y_1 + y_2 + y_4 + y_5) + 2(y_3)] \\&= \frac{3 \times 0.2}{8} [1 + 4.220696 + 3(1.040811 + 1.173511 + 1.896481 \\&\quad + 2.718282) + 2(1.433329)] \\&= 2.14309\end{aligned}$$

Homework Problem:

Q: Approximate the value of $\int_0^1 \ln(1+x) dx$

Use Simpson's $3/8$ rule for first 3 intervals and Simpson's $1/3$ rule for last 2 intervals. Take $n=5$

Solution :

$$\text{Let, } y = f(x) = \ln(1+x)$$

$$\therefore h = \frac{b-a}{n} = \frac{1-0}{5} = 0.2$$

We have, $f(x) = \ln(1+x)$

x	0.0	0.2	0.4	0.6	0.8	1.0
$f(x)$	0.0000	0.18232	0.336472	0.470003	0.587786	0.6931471

Now, applying Simpson's $\frac{3}{8}$ on $[x_0, x_3]$ - 3 interval

$$I_1 = \frac{3h}{8} [y_0 + y_3 + 3(y_1 + y_2)]$$

$$= \frac{3 \times 0.2}{8} [0 + 0.470003 + 3(0.18232 + 0.336472)]$$

$$= 0.1519784$$

Applying Simpson's $\frac{1}{3}$ on $[x_3, x_5]$ - last 2 interval.

$$I_2 = \frac{h}{3} [y_3 + 4y_4 + y_5]$$

$$= \frac{0.2}{3} [0.470003 + 4(0.587786) + 0.6931471]$$

$$= 0.23428627$$

$\therefore$ Total Approximation,

$$I = I_1 + I_2$$

$$= 0.1519784 + 0.23428627$$

$$= 0.386264673 \text{ ans.}$$

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